EE-312L Signals and Systems Lab
Laboratory Session 14
To Understand and Express the Partial Fraction Expansion of a Rational Function using MATLAB

January 1, 2025

1 Objective

The objective of this lab session is to understand and express the partial fraction expansion of rational function using MATLAB.

2 Partial Fraction Expansion of Rational Function

In the usual case, the Laplace transform of a function is expressed as a rational function of s, that is, is given as a ratio of two polynomials of s.

$$X(s) = \frac{B(s)}{A(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \ldots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0} \tag{1}$$

The MATLAB command residue(B,A) is used to compute roots of a function, where

- $B = [b_m, \ldots, b_0]$, vector of the coefficients of the numerator polynomial,
- $A = [a_n, \ldots, a_0]$, vector of the coefficients of the denominator polynomial.

After executing the command [R,P,K] = residue(B,A), the signal $X(s)$ can be written in partial fraction form as

$$X(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \ldots + \frac{r_n}{s - p_n} + K, \tag{2}$$

where

- $R = [r_1, r_2, \ldots, r_n]$

- $P = [p_1, p_2, \ldots, p_n]$
- K is the residue of the division of the two polynomials
- K is null when $m < n$

Example 01:

Express in partial fraction form the signal that in the s-domain is given by

$$X(s) = \frac{B(s)}{A(s)} = \frac{s^2 + 3s + 1}{s^3 + 5s^2 + 2s - 8} \tag{3}$$

in the frequency interval $0 \leq \omega \leq 10$ rad/s.

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clc;
clear all;
close all;

num = [ 1 3 1];
den = [ 1 5 2 -8];
[R,P,K] = residue (num,den)

syms s
X = R(1)/(s-P(1)) + R(2)/(s-P(2))+ R(3)/(s-P(3))
pretty(X)
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Results:

R =
0.5000
0.1667
0.3333
P =
-4
-2
1

$$\frac{1}{2(s+4)} + \frac{1}{6(s+2)} + \frac{1}{3(s-1)}$$

3 Lab 14: Lab Tasks

Express the following in partial fraction form

1. $$X(s) = \frac{s^2 + 3s + 1}{s^3 - 3s + 2} \tag{4}$$

2. $$X(s) = \frac{s^2 - 12s + 11}{s^2 + 4s + 3} \tag{5}$$

3. $$X(s) = \frac{s^3 - 3s + 2}{s^2 + 4s + 5} \tag{6}$$

References